Dear Students,

Blockworks just announced $100 in free API Credits.

https://app.blockworks.com/claim

Group 1: Aave Analysis / Price model

1) What "industry" is Aave a part of, what are its competitors?
    a. How does Aave's market share stack up (use TVL statistics).
    b. Competitors include : Compound, Dolomite, World Liberty Financial, Kamino
2) What are Aave's most successful products (ie, which lending markets)
    a. Which products produce the most revenue for Aave? What is the breakdown?
        i. What about their Tokenized Stocks on Base?
    b. Which blockchains does most of the revenue come from?
    c. Don't forget Flashloans
    d. Don't forget the GHO Stablecoin
    e. Don't Forget Umbrella Staking
3) Analyze the Aave Token itself
    a. What powers does the token give to users?
        i. What can you do with Aave Tokens? (ie, can you borrow against them, etc?)
    b. Does Aave do Buybacks? How are they executed?
    c. Where is the best place to trade Aave?
        i. Which CEX's
        ii. What funding rates?
        iii. Which DEX's have the deepest liquidity?
            1. How much Aave could you buy?
    d. Who are the largest holders of Aave token?
4) Can you find any interesting correlations or relationships that explain Aave's token price?
    a. Possible statistics to investigate
        i. TVL
        ii. Utilization Rate (ie, reserve ratio)
        iii. Borrow rate, USDC/USDT
        iv. Borrow Rate, ETH/BTC
        v. BTC/ETH returns/ volume.

vi. Aave Treasury: https://etherscan.io/address/0x464C71f6c2F760DdA6093dCB91C24c39e5d6e18c

5) Can you analyze how effective changes in the borrow rate are at bringing utilization back down? ie, how quicky do users react to changes in interest rates?
6) Can you analyze aave users, and find what most are using it for? For example. X% of users just deposit to earn yield, Y% of users deposit to fund leveraged strategies, Z% of users deposit to go long/short, or engage in leveraged basis trading.

This paper references AAVE $ interest rates as the "risk free rate" of crypto. Read the paper for Ideas on where you can take this analysis.

https://www.bis.org/publications/working-paper-1382-are-stablecoins-and-bank-deposits-substitutes.pdf

Another old paper that mentions aave.

https://www.bis.org/publications/working-paper-1183-why-defi-lending-evidence-aave-v2.pdf

Recent Tweet about "Humpy," a big Aave user.

https://x.com/0xGeeGee/status/2117457877569016178?a=d18TtY30IQrk5ahLV2M&s=20

Group 2: Hyperliquid:

Recent Links:

Stanley Druckenmiller logs in to Twitter to mention Trade.xyz's deal with Hyperliquid

https://x.com/standuquesne/status/2115777839622689036?a=haa7Z7sBL3RAjfuYYdM&s=20

TODAY CFTC approves single-stocks Perps in the US.
https://x.com/CFTC/status/2117618434012070013?a=1kNUi-MvQUOF2BX5dWs&s=20

Hyperliquid's borrowing market is very large and growing
https://x.com/HyperliquidNews/status/2115833774248505833?a=QXssb08OQ54BMpOEMyo&s=20

1) What are hyperliquid's revenue sources and breakdown each (Spot, Perps, HLP-3, HLP4, Portfolio Margin, USDC yield Buybacks)
    a. Which products provide what % of revenue?
    b. Can you extrapolate a growth rate?

2) HLP3 and HLP 4 require 500,000 HYPE staked to deploy markets – how much HYPE has been staked in this way? Can we expect future deployers to also buy 500,000 hype to do this?
3) HYPE token is also used as a Staking token for Hyperliquid Validators, much of HYPE is staked in this way, what is the staking yield and where does it come from?
    a. Much like Lido, Hyperliquid has a few Liquid Staking Protocols. How much hyperliquid is being staked via these protocols, versus natively.
4) Holding Hyperliquid token allows users to get discounts on their maker/taker fees when trading – How many users take advantage of these features?
5) Hyperliquid gets paid for every $ of volume that flows through its books. What are the trends re volume, and how effective is volume at predicting fees and the HYPE price?
6) HYPE token itself can be traded on Hyperliquid, Analyze the historical volume and funding rates and see if you can find a pattern. Is volume correlated with USDC deposited into Hyperliquid?
7) Finally, make a valuation for HYPE based on your assumptions re volume, fee capture etc. You will also need to assume some percentage of the total market for trading single stocks – and what would the valuation of HYPE be if they captured X % of that market.
8) Also, specify a historical regression to find the historical drivers of the HYPE price. How is this valuation different from the one you generated in 7?

Group 3: CARDS

Some very cool news about CARDS recently.

https://x.com/JamesonCamp/status/2117090283745853721?a=cToOwAkbDGA2qp39cRA& (not about CARDS specifically, but about their business model)

https://x.com/Kunallegendd/status/2117218569951818063?a=m-BvrBNZ0PiLybh5io8&s=20

https://x.com/Kunallegendd/status/2116356470258422163?a=-DGmU5kjjBP30L2qEPk&s=20 (very cool, they are making an Index for Pokemon cards. If you can get access to this pricefeed you can use it for correllations etc.

https://x.com/Collector_Crypt/status/2117665949348409531?a=NDF0vb17Sy8tzfnZ06I&s=20

1. What is their product?

    a. Gacha, Market, explain how they work, how CARDS generated revenue
    b. What percent of people automatically sell it back to CARDS, like it says in tweet 1?
    c. How many DAUs, user growth?
    d. How many CARDS in their vaults?
    e. Which cards generate the most sales/revenues?
    f. Can you collect statistics re TVL, Volume, Revenue etc and try to extrapolate it out? Ie, so we can value CARDs.
2. Competitors, are there any similar sites?
3. How does the insurance work? What percent are insured? Who is insuring it? Does CARDS generate any revenue from it?\
4. Explain the Tokenomics of CARDS, how the buybacks work etc.
    a. https://solscan.io/account/3PnVBrb4wPLFLW38oaYR7dA6HSfKpawxHGESPj5kF1QB (treasury wallet)
    b. https://solscan.io/account/3nGNwiz1qevPjhEoQi1dLj16oTkjmbevTnpV9piWY7Kq (buyback wallet)
5. Who are the owners of CARDS?
    a. What is Metaplex?
    b. How did the airdrop work? Did these wallets sell CARDS?
    c. Does owning CARDS entitle you to any rights, besides fees/buybacks?
6. Analyze CARDS liquidity
    a. Where is the deepest liquidity, around what prices? Who are the liquidity providers?
    b. Can you trade CARDS on other venues, like Hyperliquid? Are there patterns in the funding rate of cards?
    c. Analyze the TVL, volume of CARDS.
7. Based on reasonable assumptions re user growth, revenue growth, cards growth etc, can you put a valuation on the Token?
8. Can you do a historical empirical analysis of the CARD token price, and explain how it has traded in the past? How does this multifactor model valuation differ from your valuation arrived at in 7?

GROUP 5: Prediction markets:

BIG SURPRISE – I recommend doing Hyperliquid Again. They have prediction markets too, under HLP-4.

https://x.com/x256xx/status/2117296986500337842?a=6wtellZIsudr9fWJu4E&s=20

Unfortunately, KALSHI and Polymarket do not have tokens, so you cannot do what other students above will do.

1) Analyze the most popular prediction markets on Kalshi, Polymarket, and Hyperliquid.
    a. Which markets have the most volume on each respective platform
    b. What are the various fees on each platform?
2) Are there any competitive markets, ie markets for the same thing on Kalshi, Poly, and Hype? That would be an excellent case study. For example, when a large trade enters Polymarket and changes the odds there, how long until the odds change on Kalshi/Hype? Ie, are there arbitrageurs in these markets?
3) Polymarket/Kalshi were made fun of because their prediction markets were VERY wrong about the Bolsonaro Election (ie, they said 20% chance win, and he won). Possibly you can analyze this
4) The most obvious use for prediction markets is politics/betting. Kalshi/Polymarket have had markets for the outcome of the midterm elections in the senate/congress for a long time, so you can see how the odds changed. Can you analyze WHY they were changing? Are there specific events they are correlated with? When one of the venues odds change, how fast for the next one to change too? Etc
5) I understand that Polymarket pays users interest on their USDC. Does Kalshi do this? Hype?

GROUP 6: Tokenization/RWA

This group needs to narrow the scope of the project significantly. You must choose. Either Robinhood tokenized stocks, Coinbase Tokenized Stocks, Ondo Tokenized Stocks, xStocks etc.

Alternatively, you can choose a SINGLE tokenized stock, and do a deep dive on it across all issuers (ie. SpaceX: don't do Anthropic, cuz that's another group). Ie, what is the distribution of the tokenized stock across blockchains/wallets. Where can you trade it onchain? What type of liquidity is there in the tokenized stock? Can you compare the tokenized volume with the NYSE/Nasdaq volume? What is the largest trade size that makes sense to do in the tokenized stock, before it would be better to just buy the stock on the Nasdaq?

https://stonkfun.xyz/ is the meme platform that allows you to launch memecoins alongside certain tokenized stocks. These memecoins are essentially marketing for the tokenized stocks, since the only way to get the memecoins is to buy the tokenized stock. This pushes up the price of the tokenized stock, incentivizing a market maker to tokenize another unit of

the stock to capture the arbitrage, which increases the total supply of a given tokenized stock. Can you correlate changes in tokenized stock prices to changes in their total supply?

You could also just analyze stonkfun, its tokenomics, and see which tokenized stocks are the most popular to pair with? Remember the BONER/HIMS stuff we discussed – you could even do a deep dive on that particular pair if you wish.

Please don't forget that aave v4 on base allows you to borrow against tokenized stocks

Group 7: Compare and contrast Perp Dexes

Choose: Aster, Lighter, Hyperliquid

Compare and contrast each venues primary markets/volume. Do they compete with one another, or do they have their own niches?

Compare and contrast their different fee structures for trades.

How does each venue differ in its calculation of funding rates?

Are there correlations between funding rates for different assets?

Which venue gives you the "best execution" for a given trade of a given size. Ie, "if I do a 10x long on Spacex it costs ___ on hyperliquid to open, and X funding rate to keep open" whereas the same trade on Lighter would cost ___ and Y rate etc."

EACH OF THESE PROJECTS HAS TOKENS (ASTER, LIT, HYPE). So you can analyze the 3 tokens as well – their holders, liquidity on DEXes etc. You can also analyze the FUNDING RATES for each token on each venue. ie, what does it cost to trade LIT on Hyperliquid? Hype on Lit? etc. Are some venues better than others?

Group 8: Hyperliquid HLP analysis.

So HLP-3 and HLP-4 are run by deployers. The crypto perpetuals are served by HLP liquidity, ie when you trade on HLP-3, your counterparty is a wall street market maker running a high frequency trading strategy. When you trade on perpetuals, you are trading against HLP and other vaults. "he house".

How does HLP decide what positions to open? Does it automatically take the opposite of customer trades?

Get historical data on HLP, as well as on the most popular "Vaults". How has HLP performed since inception versus the other vaults. Which vaults saw the best returns, which saw the worst? https://app.hyperliquid.xyz/vaults idk how to get this data.

These vault "returns" do not account for how much RISK was taken by the vault. It would be VERY interesting if we could come up with an estimate for the RISK taken by each vault, so we could calculate Sharpe Ratios for the vaults, and evaluate their skill. One easy way to do this is just to calculate the variance of the returns of each vault.

Group 9: USDT/USDC

Interesting Tweet regarding total Stablecoin Supply:

https://x.com/matthew_sigel/status/2117690578645508511?a=vuMrMzgC4on-5M3xei8&s=20

It looks like Bloomberg has data on Stablecoin supply (based on the picture in the tweet)

The "total stablecoin supply" is an interesting statistic. Its used as a proxy for the "money supply" of crypto, like how monetary economists use aggregates like M1, M2 etc and correlate them with GDP, inflation, stock market cap etc.

Check out this paper re Stablecoins. https://www.bis.org/publications/working-paper-1382-are-stablecoins-and-bank-deposits-substitutes.pdf

The US bankers are very worried that stablecoins will cause deposits to leave US banks, cuz Stablecoins can earn higher yields than banks. In this paper, the BIS shows that there is some relationship between stablecoins supply and changes in interest rates.

You could extend their analysis to OTHER interest rates BESIDES bank deposits. ie, AAVE supply rates. They used https://defirateindex.com/

1. Obviously, you need to get LOTS of data to report on USDT and USDC correctly. USDT exists on A LOT of chains. https://tether.to/en/transparency/?tab=usdt. I am not sure if bloomberg reports the correct consolidated USDT supply.
2. We are interested in understanding HOW people are using stablecoins.
    a. Do they just hold them in their wallet like savings? What percentage of their balance is moved every day? Month? Year?
    b. Descriptive statistics on usage: Avg/Median Holding period, Aggregate turnover rate,
3. Compare 2 different large liquidity pools, like Uniswap ETH/USDC .01% and ETH/USDT .01%, are there substantial differences in these pools usage? Which

pools see more volume? Theoretically, there is an arbitrage between these two pools, since users can always swap USDC for USDT.

   a. What is the USDC/USDT volume? Is there a trend? Why would someone want to swap from one to the other?

4. rlUSD is another famous Stablecoin, issued by Ripple. Except nobody actually uses Ripple. You can analyze the descriptive statistics on rlUSD, and I expect you will find that rlUSD is SUBSTANTIALLY different from USDC/USDT.

Group 10: Anthropic Pre-IPO case study:

Need ALL the data re volume, open positions etc. for each of the HLP-3 Anthropic markets, possibly you can also examine Anthropic markets on Binance, Lighter, or Aster (if they exist). Need an account-level analysis of every trader's position and their current profit/loss.

Somewhat hard to get this data probably, but please find it, or else do a different project.

Group 12: Flashbots/MEV.

Flashbots is pretty straight forward, idk where to find data on it though, here is some.

https://collective.flashbots.net/t/flashbots-relay-bulk-data-exports-payloads-and-builder-submissions/700

https://tokenterminal.com/explorer/projects/flashbots/metrics/fees

That said, You may have noticed that Binance was Hacked this week for 6.3 billion.

https://x.com/spreekaway/status/2115904721856651382?a=B4BuOUNut6ID6Pc20EQ&s=20

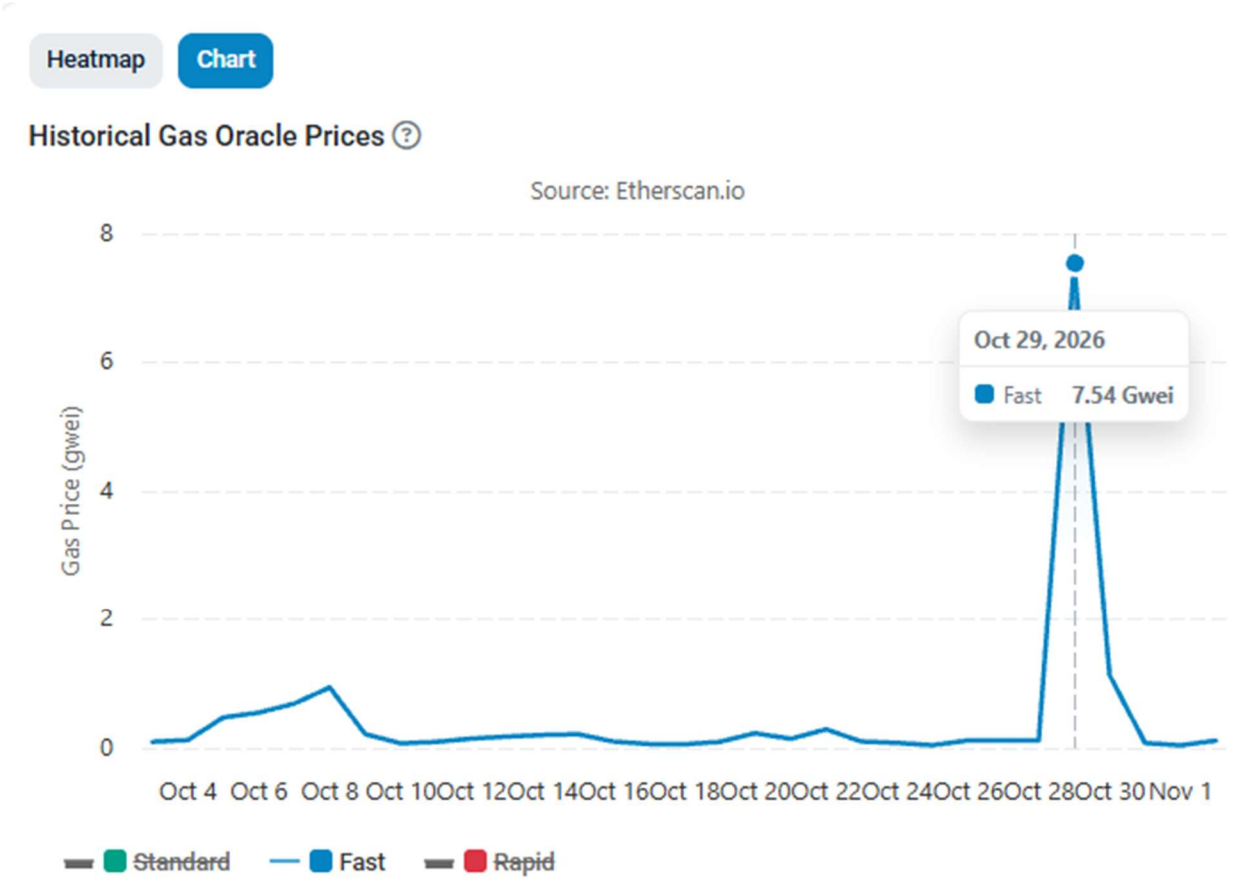
$2 Billion in stablecoins was frozen, and the hacker still got away with 4 billion or so in tokens. However, the hacker created A LOT of arbitrage/MEV opportunities as a result of their activities. Check out this one: https://x.com/Millon_Sancet/status/2116402054759645655?a=Zr3ZXWM4li67E2NtQow&s=20

And my reply tweet: https://x.com/ReidTymcio/status/2116605231496610148?a=ZPG6ow-Dt3CIruo92xo&s=20

It would be AMAZING If you could evaluate this hypothesis. Ie, how much of the 4.3 billion or so stolen was lost to MEV?

Here is a good place to start:
https://x.com/dcfgod/status/2115919171179704632?a=tp6kY9WdvEhWiDivRCs&s=20

The hackers were trying to sell their USDC/USDT cuz it can be frozen, so they bought a ton of ETH on DEXes, causing the ETH price on DEX to go up to 3000, while the real price was actually 2200. This caused an arbitrage opportunity where MEV bots, and arbitrage bots were able to earn a free $800 per ETH sold, and this caused the gas price to increase dramatically, which caused Ethereum Miners to make lots of money on fees.



So lots of people made money from the hack. Can you calculate exactly how much money was lost by the hacker to MEV?

Here is Binance's own explanation of where the money went.
https://binance.lamps.works/?tab=breakdown

Etherscan has labeled the thief wallets as “Binance Hot Wallet Exploiter 1-16”. So the procedure here is to find each one of these wallets, scan the transactions, and see how many sandwich attacks you can find. Also calculate how much the hacker overpaid for ETH, cuz that amount at minimum was earned by arbitrageurs.

This is a very relevant topic, so I’m sure people would love to see your analysis.